



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 7a (Summary and Tutorial)

Topic: Vectors 2 (Lines and Planes)

Summary for Lines and Planes

Equation of a Line: $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ where

$\mathbf{r}$ is the position vector of a general point on the line,

$\mathbf{a}$ is the position vector of a known point on the line and

$\mathbf{b}$ is the vector that indicates the direction parallel to the line (known as the *direction vector*)

Note : $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ is **not unique**. There are many ways to represent a line.

Different Forms of Equations of a Line

$$\mathbf{r} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \lambda \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \lambda \in \mathbb{R} \quad \text{– Vector Form}$$

$$\Leftrightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \lambda \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \lambda \in \mathbb{R}$$

$$\Leftrightarrow x = a_1 + \lambda b_1, y = a_2 + \lambda b_2, z = a_3 + \lambda b_3 \quad \text{– Parametric Form}$$

$$\Leftrightarrow \text{If } b_1 \neq 0, b_2 \neq 0 \text{ and } b_3 \neq 0,$$

$$\frac{x-a_1}{b_1} = \lambda, \frac{y-a_2}{b_2} = \lambda, \frac{z-a_3}{b_3} = \lambda$$

$$\Leftrightarrow \frac{x-a_1}{b_1} = \frac{y-a_2}{b_2} = \frac{z-a_3}{b_3} \quad \text{– Cartesian Form}$$

Equation of a Plane: $\mathbf{r} \cdot \mathbf{n} = d$ where

$\mathbf{r}$ is the position vector of a general point on the plane,

$\mathbf{n}$ is the normal vector to the plane (a vector perpendicular to the plane)

d is a scalar constant.

Notes:

Any point A on the plane, with position vector $\mathbf{a}$, will always give the result, $\mathbf{a} \cdot \mathbf{n} = d$

The vector equation of the plane can always be reduced to the form $\mathbf{r} \cdot \hat{\mathbf{n}} = D$. In this

form, the normal vector is reduced to a unit vector and $|D|$ is the *shortest distance of the plane from the origin*.

Different Forms of Equations of a Plane

Scalar Product form: $\mathbf{r} \cdot \mathbf{n} = d$

Cartesian form: $n_1x + n_2y + n_3z = d$ where $\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}$

Vector form: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{m}_1 + \mu\mathbf{m}_2$, $\lambda, \mu \in \mathbb{R}$

$\mathbf{a}$ is the position vector of a point on the plane,

$\mathbf{m}_1$ and $\mathbf{m}_2$ are non-parallel vectors that are parallel to the plane.

Thus, $\mathbf{m}_1 \times \mathbf{m}_2$ will be a normal to the plane.

We will now summarize the various relationships involving points, lines and planes.

	Point	Line	Plane
Point	1) Distance between points 2) Collinearity between 3 points	1) Point on line (verification) 2) Point not on line: a. Distance between point and line b. Foot of perpendicular from point to line c. Reflection of point in line 3) Length of projection of line segment on line	1) Point on plane (verification) 2) Point not on plane: a. Distance between point and plane b. Foot of Perpendicular from point to plane c. Reflection of point in plane 3) Length of projection of line segment on plane
Line		1) Angle between 2 lines 2) Relation between 2 lines: a. Parallel lines • Shortest distance between 2 lines b. Non-parallel lines with point of intersection c. Skew Lines	1) Relation between line and plane a. Line parallel to plane but not on plane b. Line lies on plane c. Line intersects plane at a point 2) Angle between line and plane 3) Reflection of line in plane (for 1a, 1c)
Plane			1) Angle between 2 planes 2) Relation between 2 planes: a. Parallel planes b. Non-parallel planes: line of intersection between planes

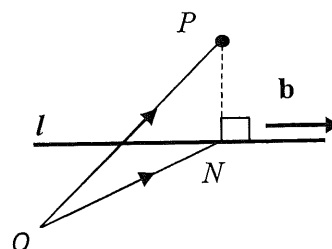
Involving Points and Lines

- ❖ To check whether the point P with position vector $\mathbf{p}$ lies on the given line l : $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, $\lambda \in \mathbb{R}$

- Let $\mathbf{p} = \mathbf{a} + \lambda\mathbf{b}$ and find a possible value for λ .
- If there is a unique value for λ from the three possible equations, then the point P lies on the line.

- ❖ To find the position vector of the foot of the perpendicular from a point P with position vector $\mathbf{p}$ to a given line l : $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, $\lambda \in \mathbb{R}$

Let N be the foot of the perpendicular from the point P to the line, l . Make use of the fact that $\overrightarrow{PN}$ is perpendicular to l



Step 1: Since N lies on l , $\overrightarrow{ON} = \mathbf{a} + \lambda\mathbf{b}$ for a unique value of λ .

Step 2: Find $\overrightarrow{PN} = \overrightarrow{ON} - \overrightarrow{OP}$ in terms of λ , where $\overrightarrow{OP} = \mathbf{p}$ is given.

Step 3: Solve for the value of λ using the fact that $\overrightarrow{PN} \cdot \mathbf{b} = 0$

Step 4: Using the value of λ found above, the position vector of N is given by $\mathbf{a} + \lambda\mathbf{b}$.

- ❖ To find the perpendicular distance (shortest distance) from a point, P with position vector, $\mathbf{p}$, to a given line l : $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, $\lambda \in \mathbb{R}$

Let N be the foot of the perpendicular from the point P to the line, l

Method 1

Step 1 : Find the position vector of the foot of the perpendicular from the point P to the given line l .

Step 2 : Perpendicular distance from the point P to the given line l is $|\overrightarrow{PN}|$.

Method 2 (Cross Product)

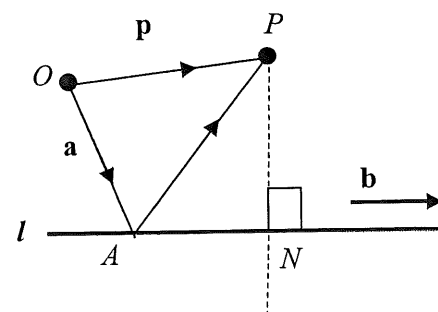
Perpendicular distance from the point P to the given line l is $|\overrightarrow{PN}| = |\overrightarrow{AP} \times \hat{\mathbf{b}}|$

Method 3 (Dot Product)

Step 1 : Find $|\overrightarrow{AN}| = |\overrightarrow{AP} \cdot \hat{\mathbf{b}}|$

Step 2 : Using Pythagoras' Theorem,
Perpendicular distance from the point P to the

given line l is $|\overrightarrow{PN}| = \sqrt{|\overrightarrow{AP}|^2 - |\overrightarrow{AN}|^2}$



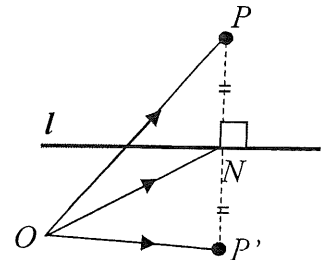
- ❖ To find the position vector of the reflection of a point, P , about a line l :
 $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$

Let P' be the reflection of P about l and that N be the foot of the perpendicular from P to l .

Method

- Find the vector $\overrightarrow{PN}$ (Note that $\overrightarrow{PN} = \overrightarrow{NP'}$)
- Position vector of $P' = \overrightarrow{OP'} = \overrightarrow{OP} + \overrightarrow{PN} + \overrightarrow{NP'} = \overrightarrow{OP} + 2\overrightarrow{PN}$

$$(\text{OR use Ratio Theorem: } \overrightarrow{ON} = \frac{\overrightarrow{OP} + \overrightarrow{OP'}}{2})$$



Involving Lines

- ❖ To check if the lines $l_1 : \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ and $l_2 : \mathbf{r} = \mathbf{c} + \mu \mathbf{d}$, $\mu \in \mathbb{R}$ are parallel

If $\mathbf{b} \parallel \mathbf{d}$ (i.e. $\mathbf{b} = k\mathbf{d}$), then $l_1 \parallel l_2$

- ❖ To find the acute angle between 2 lines, $l_1 : \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ and $l_2 : \mathbf{r} = \mathbf{c} + \mu \mathbf{d}$, $\mu \in \mathbb{R}$

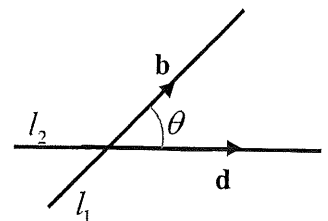
The acute angle between the lines is determined by the two direction vectors $\mathbf{b}$ and $\mathbf{d}$.

Method

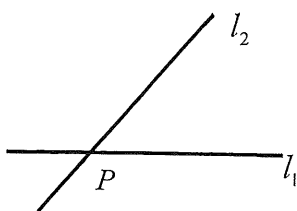
Make use of scalar product of the two direction vectors $\mathbf{b}$ and $\mathbf{d}$.

- $|\mathbf{b} \cdot \mathbf{d}| = |\mathbf{b}| |\mathbf{d}| \cos \theta$ where θ is the acute angle between the 2 lines (equivalent to angle between direction vectors, $\mathbf{b}$ and $\mathbf{d}$, if $\mathbf{b} \cdot \mathbf{d} > 0$, or angle between $-\mathbf{b}$ and $\mathbf{d}$ or $\mathbf{b}$ and $-\mathbf{d}$, if $\mathbf{b} \cdot \mathbf{d} < 0$).

$$\text{Therefore } \cos \theta = \frac{|\mathbf{b} \cdot \mathbf{d}|}{|\mathbf{b}| |\mathbf{d}|}$$



- ❖ To find the position vector of the point of intersection of the two lines $l_1 : \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ and $l_2 : \mathbf{r} = \mathbf{c} + \mu \mathbf{d}$, $\mu \in \mathbb{R}$



Let P be the point of intersection of the 2 lines l_1 and l_2 , and $\mathbf{p}$ be the position vector of P .

Then P lies on both lines and $\mathbf{p} = \mathbf{a} + \lambda \mathbf{b}$ and $\mathbf{p} = \mathbf{c} + \mu \mathbf{d}$ for some values of λ and μ .

Method

Step 1 : Since the lines intersect, $\mathbf{p} = \mathbf{a} + \lambda \mathbf{b} = \mathbf{c} + \mu \mathbf{d}$

Step 2 : From Step 1, we will have 3 linear equations in terms of λ and μ .

Solve for the values of λ and μ

- If there exist unique values for λ and μ , then the position vector of the intersection point is given by $\mathbf{a} + \lambda \mathbf{b}$ or $\mathbf{c} + \mu \mathbf{d}$.
- If there are no unique values for λ and μ , then the 2 lines are non-intersecting lines. (They can be either parallel or skew lines)

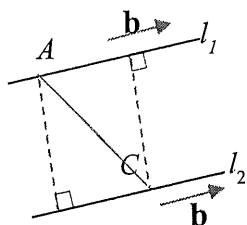
- ❖ To check if the lines $l_1 : \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ and $l_2 : \mathbf{r} = \mathbf{c} + \mu \mathbf{d}$, $\mu \in \mathbb{R}$ are skew lines (Skew lines are non-parallel and non-intersecting lines.)

Method

Step 1 : Show that l_1 and l_2 are not parallel ($\mathbf{b} \neq k \mathbf{d}$)

Step 2 : Then assume $\mathbf{a} + \lambda \mathbf{b} = \mathbf{c} + \mu \mathbf{d}$ and show that there are **NO** unique values for λ and μ that satisfy the equations formed, i.e., the 2 lines do not intersect at a common point.

- ❖ To find the shortest distance between two parallel lines $l_1 : \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ and $l_2 : \mathbf{r} = \mathbf{c} + \mu \mathbf{b}$, $\mu \in \mathbb{R}$.



The shortest distance between two parallel lines can be found by taking any point, says A on $l_1 : \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ and finding the shortest distance between point A and $l_2 : \mathbf{r} = \mathbf{c} + \mu \mathbf{b}$, $\mu \in \mathbb{R}$.

OR

The shortest distance between two parallel lines can be found by taking any point, says C on $l_2 : \mathbf{r} = \mathbf{c} + \mu \mathbf{b}$, $\mu \in \mathbb{R}$ and finding the shortest distance between point C and $l_1 : \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$.

(Refer to section on shortest distance between a point and a line)

Involving Points, Lines and Planes

- ❖ To check whether the point P with position vector $\mathbf{p}$ lies on the given plane $\pi : \mathbf{r} \cdot \mathbf{n} = d$

Method

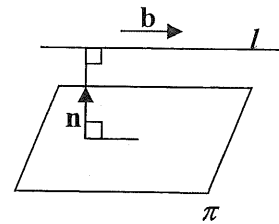
– If $\mathbf{p} \cdot \mathbf{n} = d$, then the point P lies on the plane.

If $\mathbf{p} \cdot \mathbf{n} \neq d$, then the point P does not lie on the plane.

- ❖ To check if the line $l: \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ is parallel to a given plane $\pi: \mathbf{r} \cdot \mathbf{n} = d$

Method

- If $\mathbf{b} \cdot \mathbf{n} = 0$, then the line is parallel to the plane.
(perpendicular to the normal)



- ❖ To check if the line $l: \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ lies in a given plane $\pi: \mathbf{r} \cdot \mathbf{n} = d$

Method 1

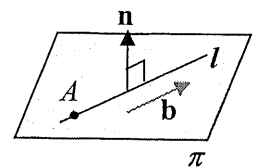
- Show that $(\mathbf{a} + \lambda \mathbf{b}) \cdot \mathbf{n} = d$ for all values of λ , then the line lies in the plane.

Method 2

- Step 1**: Show that $\mathbf{b} \cdot \mathbf{n} = 0$, then the line is parallel to the plane.

- Step 2**: Show that $\mathbf{a} \cdot \mathbf{n} = d$, then the point with position vector, $\mathbf{a}$ lies on the plane.

Then conclude the line lies in the plane.

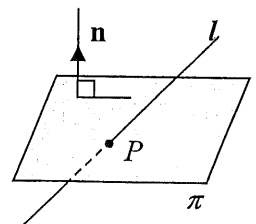


- ❖ To find the position vector of point of intersection between the line $l: \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ and the plane $\pi: \mathbf{r} \cdot \mathbf{n} = d$

Let P with position vector $\mathbf{p}$ be the point of intersection between the line l and the plane π . Then $\mathbf{p} = \mathbf{a} + \lambda \mathbf{b}$, for some $\lambda \in \mathbb{R}$ and $\mathbf{p} \cdot \mathbf{n} = d$

Method

- Since $\mathbf{p} = \mathbf{a} + \lambda \mathbf{b}$, for some $\lambda \in \mathbb{R}$ and $\mathbf{p} \cdot \mathbf{n} = d$, then $(\mathbf{a} + \lambda \mathbf{b}) \cdot \mathbf{n} = d$.
– Solve for λ and use this value to find the position vector of the point of intersection which is given by $\mathbf{p} = \mathbf{a} + \lambda \mathbf{b}$



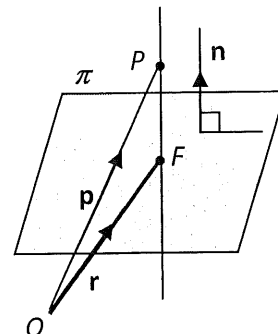
- ❖ To find the position vector of the foot of the perpendicular from the point P to the plane $\pi : \mathbf{r} \cdot \mathbf{n} = d$

Let F be the foot of the perpendicular from the point P to the plane $\pi : \mathbf{r} \cdot \mathbf{n} = d$.

Method

Step 1: Find vector equation of the line that is perpendicular to the plane (hence direction vector of this line is $\mathbf{n}$) and passing through $P \Rightarrow \mathbf{r} = \mathbf{p} + \lambda \mathbf{n}, \lambda \in \mathbb{R}$

Step 2: The point of intersection between line, $\mathbf{r} = \mathbf{p} + \lambda \mathbf{n}, \lambda \in \mathbb{R}$ and the plane $\pi : \mathbf{r} \cdot \mathbf{n} = d$ is the foot the perpendicular from the point P to the plane.



Consider $(\mathbf{p} + \lambda \mathbf{n}) \cdot \mathbf{n} = d$ and solve for λ to use this value to find the position vector of the point given by $\mathbf{p} + \lambda \mathbf{n}$.

(Same as above for the point of intersection between a line and a plane)

- ❖ To find the distance between point P with position vector $\mathbf{p}$ and the plane $\pi : \mathbf{r} \cdot \mathbf{n} = d$

Let F be the foot of the perpendicular from the point P to the plane $\pi : \mathbf{r} \cdot \mathbf{n} = d$.

Method 1

Step 1: Find the position vector of the foot of the perpendicular from the point P to the plane $\pi : \mathbf{r} \cdot \mathbf{n} = d$.

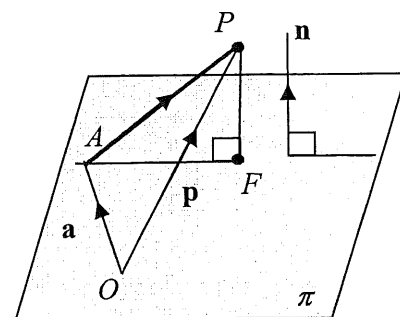
Step 2: The distance between the point and the plane is given by $|\overrightarrow{PF}|$.
(Refer to diagram above)

Method 2

Step 1: Find any point, says A on the plane with position vector $\mathbf{a}$, such that $\mathbf{a} \cdot \mathbf{n} = d$.

Step 2: Find $\overrightarrow{AP} = \mathbf{p} - \mathbf{a}$

The shortest distance between the point P and the plane is given by $\frac{|(\mathbf{p} - \mathbf{a}) \cdot \mathbf{n}|}{|\mathbf{n}|}$, that is length of projection of AP on the normal vector of the plane.



Method 3

The distance between the point and the plane is given by $\frac{|\mathbf{p} \cdot \mathbf{n} - d|}{|\mathbf{n}|}$.

Note from method 2: $\frac{|(\mathbf{p} - \mathbf{a}) \cdot \mathbf{n}|}{|\mathbf{n}|} = \frac{|\mathbf{p} \cdot \mathbf{n} - \mathbf{a} \cdot \mathbf{n}|}{|\mathbf{n}|} = \frac{|\mathbf{p} \cdot \mathbf{n} - d|}{|\mathbf{n}|}$.

- ❖ To find the acute angle between the line $l: \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$ and the plane $\pi: \mathbf{r} \cdot \mathbf{n} = d$

The desired angle is θ .

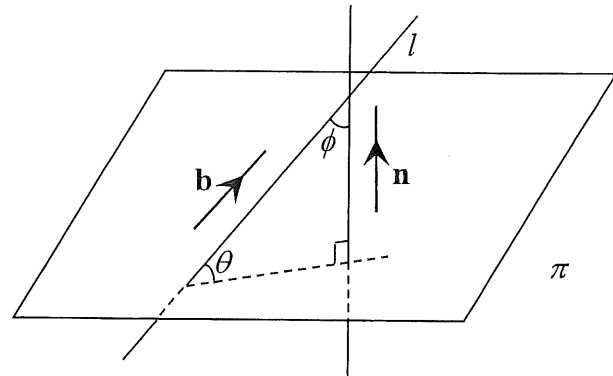
Let the acute angle between $\mathbf{n}$ and $\mathbf{b}$ be ϕ .

Method 1

Step 1: Find the acute angle between $\mathbf{n}$ and $\mathbf{b}$

$$\text{using } \cos \phi = \frac{|\mathbf{n} \cdot \mathbf{b}|}{|\mathbf{n}| |\mathbf{b}|}$$

Step 2: The acute angle between the line and the plane is $\theta = 90^\circ - \phi$.



Let the acute angle between the line and the plane be θ . Then $\phi = 90^\circ - \theta$.

Method 2

We can also use a single step to get θ using $\sin \theta = \frac{|\mathbf{n} \cdot \mathbf{b}|}{|\mathbf{n}| |\mathbf{b}|}$.

(since $\cos \phi = \cos (90^\circ - \theta) = \sin \theta$)

• **Involving Planes**

- ❖ To check if the two planes $\pi_1: \mathbf{r} \cdot \mathbf{n}_1 = d_1$ and $\pi_2: \mathbf{r} \cdot \mathbf{n}_2 = d_2$ are parallel.

Method

– If $\mathbf{n}_1 \parallel \mathbf{n}_2$ (i.e. $\mathbf{n}_1 = k \mathbf{n}_2$), then $\pi_1 \parallel \pi_2$

- ❖ To find the distance between two parallel planes $\pi_1: \mathbf{r} \cdot \mathbf{n}_1 = d_1$ and $\pi_2: \mathbf{r} \cdot \mathbf{n}_2 = d_2$

Method

Step 1: Find any point, says A with position vector, $\mathbf{a}$ on the plane π_1 .

Step 2: Find the shortest distance between A and π_2 which is $\frac{|\mathbf{a} \cdot \mathbf{n}_2 - d_2|}{|\mathbf{n}_2|}$

(Refer to Distance between point and the plane $\pi: \mathbf{r} \cdot \mathbf{n} = d$)

- ❖ To find the acute angle between two planes $\pi_1: \mathbf{r} \cdot \mathbf{n}_1 = d_1$ and $\pi_2: \mathbf{r} \cdot \mathbf{n}_2 = d_2$

Let the acute angle between the two planes be θ .

Method

– To find the acute angle between the two planes, we make use of $\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|}$

❖ To find vector equation of the line of intersection between two planes $\pi_1: \mathbf{r} \cdot \mathbf{n}_1 = d_1$
and $\pi_2: \mathbf{r} \cdot \mathbf{n}_2 = d_2$

Method 1 (using GC)

Step 1: Find the cartesian equations of the planes.

Step 2: Use GC to find the line of intersection.

Method 2 (if GC is not allowed)

Step 1: Find the cartesian equations for the planes

Step 2: Let one of the variables says $x = 0$. Then solve for y and z using the cartesian equations from **Step 1** to get a common point, A with position vector, $\mathbf{a}$.

Step 3: Compute $\mathbf{b} = \mathbf{n}_1 \times \mathbf{n}_2$ which is the direction vector of the line of intersection.
Then the equation of the line of intersection is $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, $\lambda \in \mathbb{R}$

Method 3 (if GC is not allowed)

Step 1: Find the cartesian equations for the planes

Step 2: Find two common points using the cartesian equations from **Step 1** by first letting $x = 0$ and solve for y and z and then letting $y = 0$ and solve for x and z to get another common point

Step 3: Use the two points to find the direction vector of the line of intersection.

Example to illustrate Method 2 and 3:

Find the equation of the line of intersection of the two planes whose equations are

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} = 6 \text{ and } \mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = 1 \text{ respectively}$$

Method 2

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} = 6 \Rightarrow x + y - 3z = 6 \text{ ----- (1)}$$

$$\text{and } \mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = 1 \Rightarrow x + 2z = 1 \text{ -----(2)}$$

Let $z = 0$ and solve, we have $x = 1, y = 5$ and $z = 0$

$$\begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \\ -1 \end{pmatrix}$$

$$\Rightarrow \text{the required vector equation is } \mathbf{r} = \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ -1 \end{pmatrix}, \lambda \in \mathbb{R}$$

Method 3

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} = 6 \Rightarrow x + y - 3z = 6 \quad \text{----- (1) and} \quad \mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = 1 \Rightarrow x + 2z = 1 \quad \text{----- (2)}$$

Let $z = 0$ and solve, we have $x = 1, y = 5$ and $z = 0$

Let $x = 0$ and solve, we have $x = 0, y = \frac{15}{2}$ and $z = \frac{1}{2}$

$$\begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 15/2 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 1 \\ -5/2 \\ -1/2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 \\ -5 \\ -1 \end{pmatrix}$$

$$\Rightarrow \text{the required vector equation is } \mathbf{r} = \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ -1 \end{pmatrix}, \lambda \in \mathbb{R}$$

Revision Tutorial Questions

1 Source of Question: HCI 2016 JC2 CTP1Q10(a)

Three distinct points O , A and B are such that $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

- (i) Let $\overrightarrow{OP} = \mathbf{r}$ and $\mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$. By expressing $\mathbf{r}$ in terms of $\mathbf{a}$ and $\mathbf{b}$, show that the point P lies on a line parallel to $\mathbf{b}$. [2]
- (ii) In terms of $\mathbf{a}$ and $\mathbf{b}$, write down the distance from O to this line. [2]
Hence write down the distance from B to this line.
- (iii) Given that the line passes through O , what can you say about the relationship between $\mathbf{a}$ and $\mathbf{b}$? [1]

2 Source of Question: CJC 2016 JC2 CTQ2

In the rhombus $OABC$ where O is the origin, the position vectors of the points A , B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively.

- (i) Show that $(\mathbf{a} - \mathbf{c}) \cdot (\mathbf{a} + \mathbf{c}) = 0$. [2]

Another point D on AB produced is such that $AB:AD = 1:4$ and the lines OD and BC intersect at the point E .

- (ii) (a) Find $\overrightarrow{OD}$ in terms of $\mathbf{a}$ and $\mathbf{c}$, [1]

- (b) By considering the lines of OD and BC , find $\overrightarrow{OE}$ in terms of $\mathbf{a}$ and $\mathbf{c}$. [4]

Hence, find the value of $\frac{BE}{EC}$. [1]

3 Source of Question: MI 2015 Prelim P1Q7

A line l passes through the points A and B with coordinates $(0, -2, 2)$ and $(1, 0, 1)$ respectively.

- (i) Find the acute angle between $\overrightarrow{OA}$ and the line l , where O is the origin. [2]
- (ii) Hence, find the shortest distance from O to the line l , leaving your answer in exact form. [1]

4 Source of Question: MI 2015 Prelim P2 Q3

Relative to a fixed origin O , the points A , B and C have position vectors $\alpha\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $4\mathbf{i} - 2\mathbf{j}$ and $-\mathbf{i} - 7\mathbf{j} + \beta\mathbf{k}$ respectively where α and β are constants.

Given that A , B and C are collinear, show that $\alpha = 5$ and $\beta = -10$. [2]

Hence state the ratio of the area of $\triangle OBA$ to the area of $\triangle OBC$. [1]

Find the vector equation of the line l which passes through points A and B . [1]

The point P lies on OB such that $\overrightarrow{OP} = \frac{1}{3}\overrightarrow{PB}$. Q is a point on the line l such that the length of projection of $\overrightarrow{PQ}$ on the line OB is $\sqrt{5}$ units. Find the possible coordinates of Q . [7]

5 Source of Question: RI 2015 Prelim P1Q6

The line l_1 has equation

$$\mathbf{r} = (3\mathbf{i} + 2\mathbf{j} + \mathbf{k}) + \lambda(\mathbf{i} + \mathbf{j} + \mathbf{k}), \lambda \in \mathbb{R},$$

and the line l_2 passes through the origin O and point A , whose position vector is given by $\mathbf{a} = 2\mathbf{i} + \mathbf{j}$. Find the acute angle between

(i) l_1 and l_2 , [2]

(ii) l_1 and the xy -plane. [2]

A point B on l_1 is such that $OB = \sqrt{5}$.

(iii) Find the possible position vectors of B . [4]

6 Source of Question: YJC 2015 Prelim P1Q12

Relative to the origin O , two points A and B have position vectors $\mathbf{a} = 2\mathbf{i} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$ respectively.

(i) Find $\mathbf{a} \times \mathbf{b}$ and give the geometrical meaning of $|\mathbf{a} \times \mathbf{b}|$. [2]

Hence write down the area of triangle OAB . [1]

(ii) Find a vector equation of the line l passing through A and B . [2]

(iii) The perpendicular to l from the point C with position vector $-13\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ meets the line at the point M . Show that the position vector of M is $\mathbf{i} + \mathbf{j} - 2\mathbf{k}$. [3]

(iv) Find a cartesian equation of the plane containing O , A and B and the exact length of projection of $\overrightarrow{CM}$ onto this plane. [4]

(v) Find the acute angle between the line OC and the triangle OAB . [2]

7 Source of Question: CJC 2015 Promo Q6 (a)

The equation of a line l is given by $\frac{x-5}{2} = \frac{y+1}{-2} = z+1$. A point P , not lying on l , has position vector $9\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}$.

- (i) Given that Q is a point on l such that $PQ = 6\sqrt{3}$, find the possible coordinates of Q . [4]
- (ii) Hence, or otherwise, find the position vector of the foot of perpendicular from P to l . [2]

8 Source of Question: NYJC 2018 JC1 Mid-Year CT/ Q5

Referred to the origin O , the points A and B have position vectors $-3\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$ and $7\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ respectively. The line l has equation $\mathbf{r} = (1 + \lambda\alpha)\mathbf{i} + (\alpha + \lambda)\mathbf{j} + (1 - 2\lambda)\mathbf{k}$, $\lambda \in \mathbb{R}$, where α is a constant.

- (i) Determine the range of values of α such that l and line AB are skew. [5]

Given that $\alpha = 2$, find

- (ii) the length of projection of $\overline{AB}$ onto l , [2]
- (iii) the acute angle between l and line AB . [2]

9 Source of Question: PJC 2018 JC1 Mid-Year CT/ Q11

An artist decides to use light and ropes to create an art installation in a room. Points (x, y, z) are defined relative to a hook of negligible size on the wall at $(0, 0, 0)$, where units are in metres. Ropes are laid in straight lines and the thickness of ropes can be neglected.

- (i) A rope starts at the hook and goes in the direction $2\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$. It is 8 metres long and ends at point P . Find the exact coordinates of P . [2]

A rope D starts at the hook and goes in the direction $\mathbf{i} + \mathbf{j}$. Another rope E is installed which passes through points $Q(3, 1, 2)$ and $R(1, -2, a)$.

- (ii) Find the values of a for which D and E will **not** meet. [3]

After considering the artistic effects, the artist uses $a = -2$.

- (iii) The artist shines a light onto E perpendicularly onto the wall, which is the $y-z$ plane. Find the length of the shadow cast by QR onto the wall. [3]

The artist wants to add two ropes F and G . Both ropes pass through point Q and are reflection of each other about rope E . It is known that the rope F goes in the direction $10\mathbf{i} + 3\mathbf{j}$. Find the vector equation of G . [5]



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 7b (Summary and Tutorial)

Topic: Vectors 3 (Lines and Planes)

Summary for Lines and Planes

[Refer to Revision 7a]

Revision Tutorial Questions – More questions on Vectors!

Source of Question: NJC Prelim 9758/2018/02/Q2

1 The planes p_1 and p_2 , have equations $2x + 3y + 6z = 0$ and $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = 6$ respectively.

(i) Find a vector equation of the line of intersection, l , between p_1 and p_2 . [2]

The line m passes through the points $A(2, 1, 1)$ and $B(5, 4, 2)$.

(ii) Verify that A lies on p_2 . [1]

(iii) Find the coordinates of the points on m that are equidistant from planes p_1 and p_2 .

[5]

Source of Question: NJC JC2 Mid-Year CT 9758/2018/01/Q6

2 The equation of the plane p is given by $ax - 2y - 5z = 7$, where a is a real constant.

(i) Given that the line l with equation $\mathbf{r} = \begin{pmatrix} 3 \\ 5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -2 \\ 0 \end{pmatrix}$, $\lambda \in \mathbb{R}$ intersects the plane p

exactly once, find the possible values of a . [1]

Assume for the remainder of this question that $a = 3$.

(ii) Find the coordinates of the point of intersection between l and p . [3]

(iii) Find the acute angle between the line l and the plane p . [2]

(iv) Find the vector equation of the line of reflection of the line l in the plane p . [5]

Source of Question: YJC Prelim 9758/2018/01/Q9

3 The two lines l_1 and l_2 have equations $x = \frac{z}{-2} - 2, y = 0$ and $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ q \\ 2 \end{pmatrix}$

respectively, where $\mu \in \mathbb{R}$ and q is a constant. The point A has coordinates $(4, 3, k)$ where k is a constant and the two planes p_1 and p_2 have equations $4x = -8 + y - 2z$ and $y + tz = 2t$ respectively, where t is a constant.

- (i) Find a vector equation of l_1 and hence show that l_1 lies in p_1 . [3]
- (ii) Find the coordinates of the foot of perpendicular from A to p_1 . Express your answer in terms of k . [3]
- (iii) Given that the angle between l_1 and l_2 is 60° , find the possible values of q . [2]
- (iv) Given that $q = 1$ and that the point $B(1, 2, 2)$ is equidistant from l_2 and p_2 , find the possible values of t . [6]

Source of Question: SAJC JC2 Mid-Year CT 9758/2018/01/Q8(b)

4 The planes Π_1 and Π_2 have equations $x + my + 3z = 6$ and $mx + 3y + z + 10 = 0$ respectively. The point P has coordinates $(-3, -3, 5)$ and O is the origin.

- (i) Given that point P is on Π_1 , find m . Verify that point P is also on Π_2 . [2]
- (ii) Hence find a vector equation of the line of intersection between the planes Π_1 and Π_2 . [2]
- (iii) Point Q with coordinates $(-3, 0, 3)$ is a point on Π_1 . Find the shortest distance from point Q to Π_2 . [3]
- (iv) Suggest in parametric form, the equation of a plane Π_3 that contains points P and Q and is perpendicular to Π_2 . [1]

Source of Question: SRJC JC2 Mid-Year CT 9758/2018/02/Q7

5 The line l_1 and the plane p_1 have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} + t \begin{pmatrix} -1 \\ -2 \\ 3 \end{pmatrix}, \quad t \in \mathbb{R} \quad \text{and} \quad p_1: \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 2.$$

It is given that the point A has position vector $3\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$.

- (i) Find the acute angle between l_1 and p_1 . [1]
- (ii) Find the coordinates of the foot of perpendicular from the point A to p_1 . [4]

The plane p_2 and the plane p_3 have equations $-2x + z = 6$ and $\alpha x + 3y + 2z = \beta$ respectively, where $\alpha, \beta \in \mathbb{R}$. Given that the plane p_1 meets the plane p_2 at the line l_2 ,

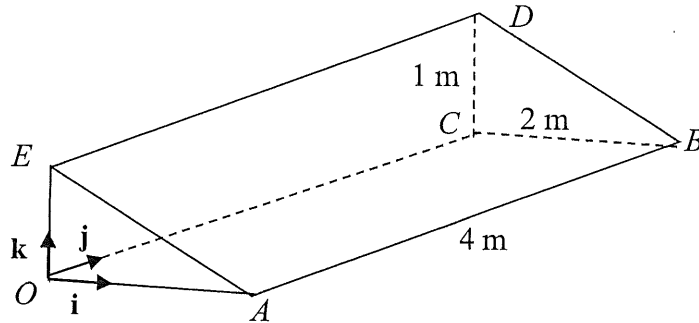
- (iii) find the equation of l_2 . [1]
- (iv) Find the values of α and β if p_3 also contains l_2 . [2]
- (v) The plane p_4 is parallel to the plane p_2 and has equation $-2x + z = k$, where $k \in \mathbb{R}$.
Find the possible values of k for which the plane p_4 is $\sqrt{5}$ units away from the plane p_2 . [2]

Source of Question: SRJC Prelim 9758/2018/02/Q5

- 6 A hollow metallic ramp, in the shape of a prism, is constructed for the marching contingent to march onto to reach an elevated platform from the ground during the national day parade.

The diagram below shows the prism with O as the origin of position vectors and the unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA , OC and OE respectively.

It is given that $OE = CD = 1$ m, $OA = CB = 2$ m and $OC = AB = ED = 4$ m.



A laser beam in the form of a line l has Cartesian equation $\frac{a+x}{2} = z, y=1$, where $a \in \mathbb{R}$, is emitted onto the plane $ABDE$.

- (i) Find, in terms of a , the coordinates of the point of intersection; M , of the laser beam and the plane $ABDE$. [4]

For the following parts of the question assume $a = 0$.

- (ii) The laser beam is reflected about the plane $ABDE$. By finding the foot of perpendicular from $Q(0,1,0)$ to the plane $ABDE$, find the equation of the reflected beam. [5]

- (iii) The path traced out by an ant crawling on the floor $OABC$ is given by

$$\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}, \beta \in \mathbb{R}. \text{ Let } P \text{ be the point on the path, located under the ramp,}$$

whereby the ant is equidistant between the planes $ABDE$ and $OCDE$. Find the position vector of point P exactly. [4]

Source of Question: TJC JC2 Mid-Year CT 9758/2018/01/Q8

7 The plane Π has equation

$$\mathbf{r} = \begin{pmatrix} -8 \\ 12 \\ 4 \end{pmatrix} + \alpha \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}, \quad \alpha, \beta \in \mathbb{R}.$$

- (i) The point A has coordinates $(1, 2, 3)$. The point B , the reflection of A in Π , has coordinates $(-5, c, -9)$. By considering the mid-point of AB , or otherwise, find the value of c . [3]

The line l is parallel to Π and has equation

$$\mathbf{r} = \begin{pmatrix} 4 \\ 0 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

- (ii) Find the distance between l and Π . [3]
- (iii) The variable point P moves along l and the variable point Q moves on Π . The point R lies on the line segment PQ such that $PR = 3RQ$.
Find $\overrightarrow{OR}$ in terms of α , β , and λ . [2]
Hence give a geometrical description of the locus of R . [2]

Source of Question: TJC JC2 Mid-Year CT 9758/2018/02/Q5

- 8 An architectural student created a cuboid model $OABCDEFG$ with dimensions 5 cm by 3 cm by 4 cm as shown in Figure 1. Due to poor construction, the cuboid leaned towards the positive x -direction to form a parallelepiped $OABCD'E'F'G'$ as shown in Figure 2 such that

- the acute angle between the plane $OCG'D'$ and the y - z plane is 30° ,
- the planes $OABC$ and $D'E'F'G'$ remain parallel to the x - y plane, and
- the planes $OAE'D'$ and $CBF'G'$ remain parallel to the x - z plane.

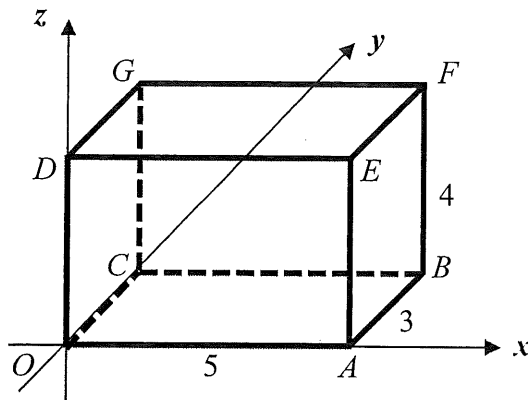


Figure 1

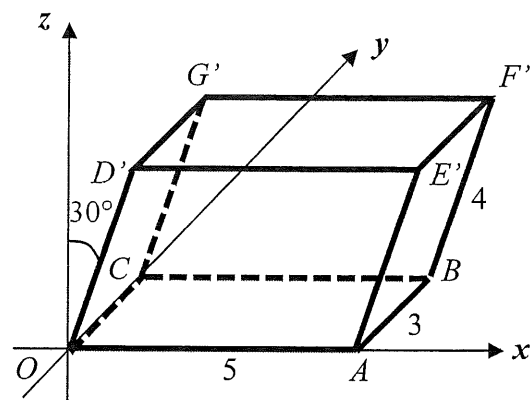


Figure 2

- (i) Show that $\overrightarrow{OD'} = 2\mathbf{i} + 2\sqrt{3}\mathbf{k}$. Hence find a vector equation of the plane $ABF'E'$ in parametric form. [3]
- (ii) Find the acute angle between the plane $ABF'E'$ and the plane with equation $\mathbf{r} \cdot \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} = 3$. [3]
- (iii) Find a vector equation of the line of intersection between the plane $ABF'E'$ and the y - z plane. [3]
- (iv) Find the position vector of the point P on OA which has a shortest distance of $\sqrt{\frac{6}{5}}$ cm from OF' . [4]